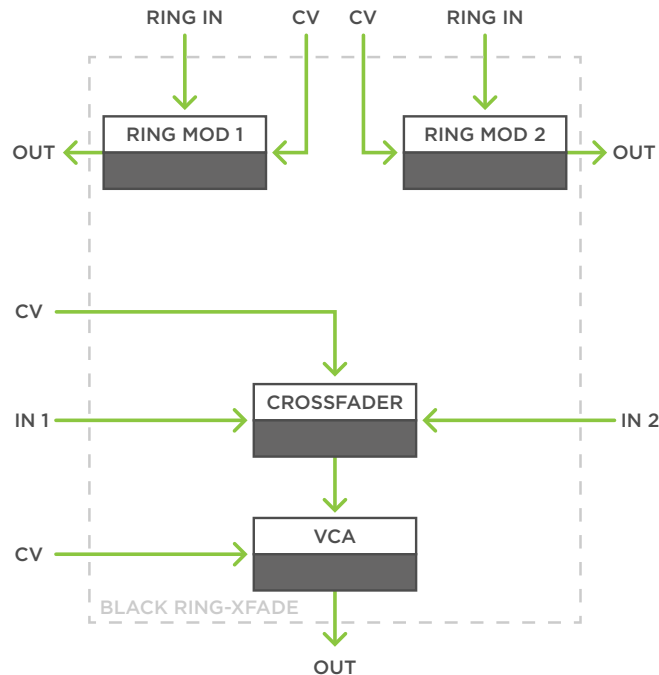


Erica Synths Black Ring-XFade

Flowchart & Internal Connections

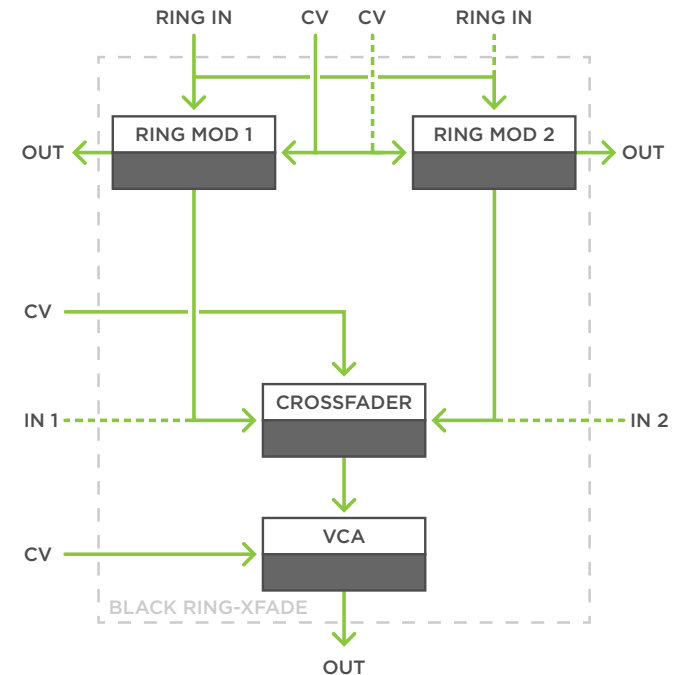
— Audio
— Control Voltage
— Trigger / Gate



(a)

a - The two ring modulators and the crossfader with build in VCA can be used as independent modules. This image shows the different sections within the Black Ring-XFade, connected to the inputs and outputs of those sections.

MONOTRAIL TECH TALK



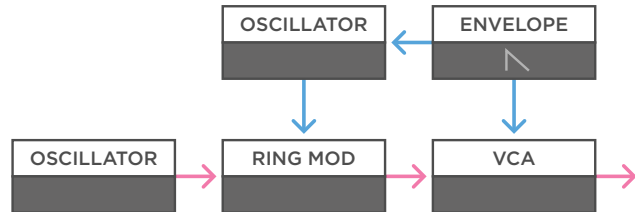
(b)

b - Here you see the internal connections, active when the dotted input lines are not used. These make it very easy to setup complex patches.

Erica Synths Black Ring-XFade

Ring Modulation

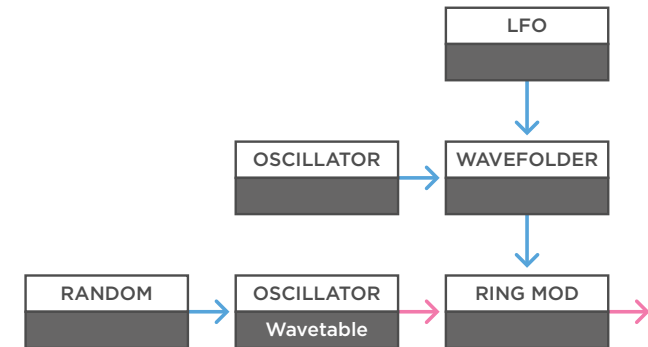
Audio
Control Voltage
Trigger / Gate



(a)

a - A classic metallic ring mod sound is achieved by using two tuned sine wave oscillators. You can feed one oscillator to the carrier input, and one oscillator to the mod input, and play around with different related tunings. For extra dynamics you can use a gate from a sequencer to trigger an envelope with a fast attack and slightly longer decay, to open a VCA at the end. And if you want, use the same or another envelope to the pitch of one of the oscillators.

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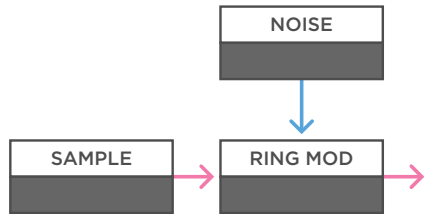
(b)

b - With ring modulators it's often interesting to modulate signals before going into the module. To make interesting drones for example, you can use a wavetable oscillator as the input, with a slow random voltage to modulate the waveshape. And use another oscillator, through a wavefolder as the modulator, again with something like a slow LFO modulating the wavefolder.

Erica Synths Black Ring-XFade

Ring Modulation

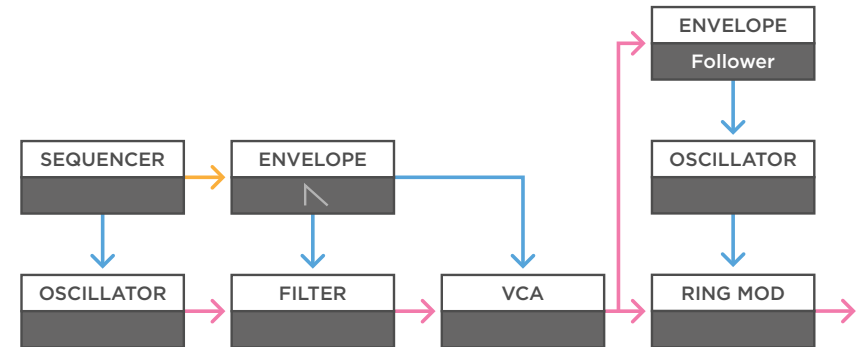
Audio
Control Voltage
Trigger / Gate



(a)

a - You can feed the ring modulator all kinds of signals, so definitely experiment with this. For example, if you use a drum loop from a sample player as the carrier, and some white noise as modulator, you can use the CV knob on the Ring-Fade, to dial in a certain amount of dust, or crunch, to the drums.

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(b)

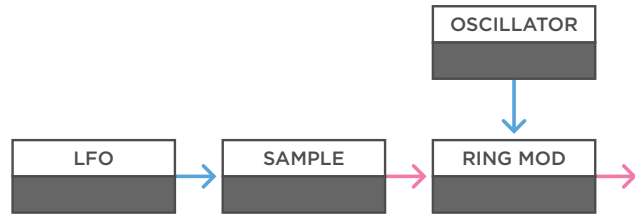
b - In this setup there is a simple voice with an oscillator, filter, VCA, and a sequencer with envelope creating a melody. The result of the entire voice is fed into the ring mod. Now you can add texture and effects by feeding different signals into the modulation input. That could be things like noise, samples, or another oscillator.

It's worth it to try and modulate the pitch of that oscillator with an envelope follower. Just multiply the signal of the voice, right before going into the ring mod, send it to the envelope follower, and then to the pitch of the oscillator.

Erica Synths Black Ring-XFade

Ring Modulation

Audio
Control Voltage
Trigger / Gate

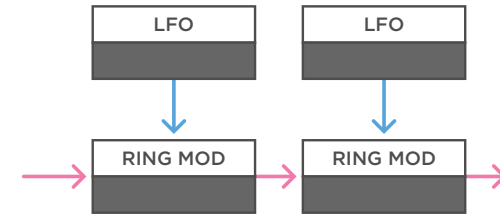


(a)

a - By experimenting with samples and field recordings, either as the carrier or modulation input, you can often come to interesting textures, and sound snippets. Try modulating samples with a simple oscillator or other samples. If you use a sample player with CV control over the playback speed, you can use a modulation source like an LFO to create extra dynamics.

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TECH TALK



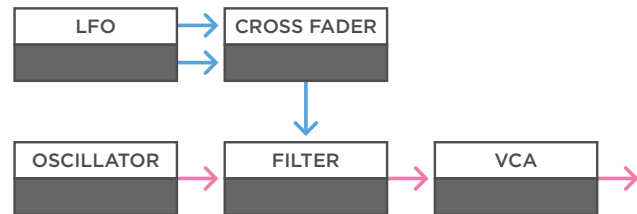
(b)

b - Because the Black Ring-Fade offers two ring modulators, you can manually patch them up in series. Just patch the output of the first, into the input of the second. Then you can experiment with using a single modulation source, which is internally multiplied to the second input, or of course, use two completely different modulators.

Erica Synths Black Ring-XFade

Crossfading

Audio
Control Voltage
Trigger / Gate

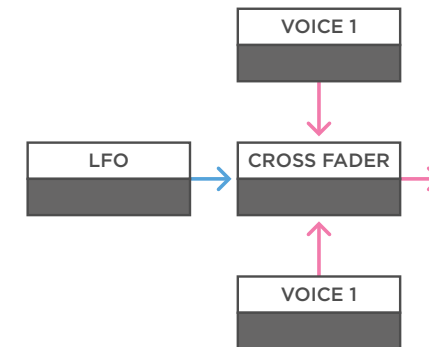


(a)

a - To create some simple wave morphing, you can feed two different waveforms of a single oscillator to a crossfader, and use something like an LFO to slowly morph between the shapes.

You can use the same trick for modulation, if you have an LFO with multiple waveform outputs.

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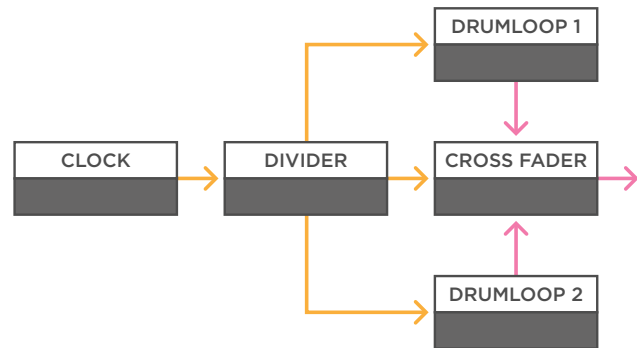
(b)

b - It's also interesting to use a crossfader to create a volume balance between two sounds. For example, if you have a large generative patch with multiple voices, you can feed the crossfader two of them, and have a single modulation source modulate the balance between the two.

Erica Synths Black Ring-XFade

Crossfading

- Audio
- Control Voltage
- Trigger / Gate

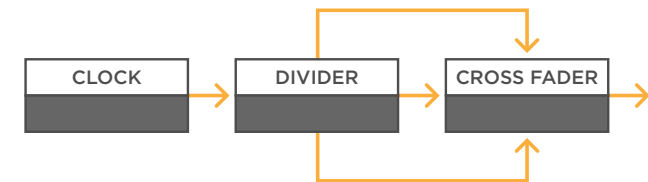


(a)

a - If you use a gate signal or square wave LFO to modulate the crossfader, you can use it to switch between two sounds. For example two different synced drum loops. In this setup you can add a clock and a clock divider to trigger the drum samples, and a division to switch between them.

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(b)

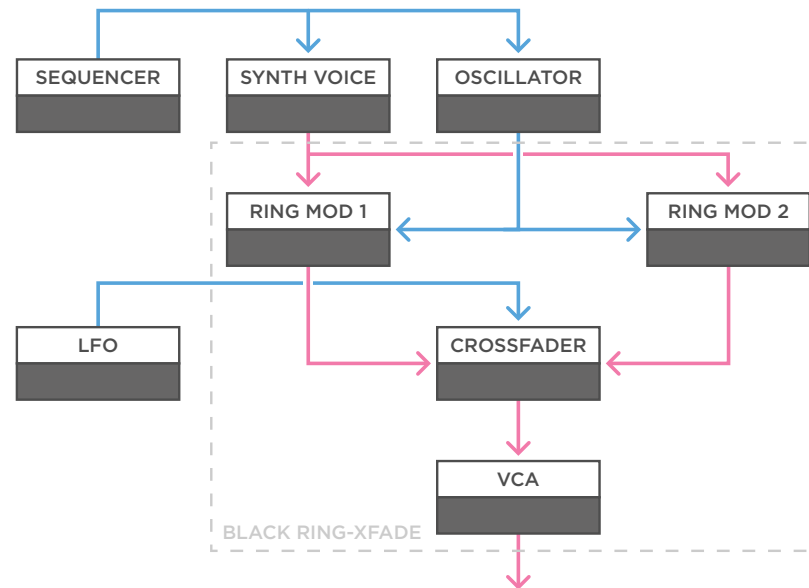
b - Similarly you can use a crossfader as a simple two input switch for the CV or triggers. For example, if you take a modular clock, and divider again, you can take two different divisions to the inputs, and use a slower division to switch between the two signals.

Erica Synths Black Ring-XFade

Combined Patches

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - In this patch multiple functions of the Ring-XFade are used. A simple synth voice with an oscillator and filter is sent to the first ring mod, and automatically multiplied to the input of the second ring mod. Then, another oscillator is used as the CV input of the first ring mod, again automatically multiplied to the CV input of the second. The output of each ring mod is internally connected to the inputs of the crossfader, and then the fade output passes through the VCA to the main output.

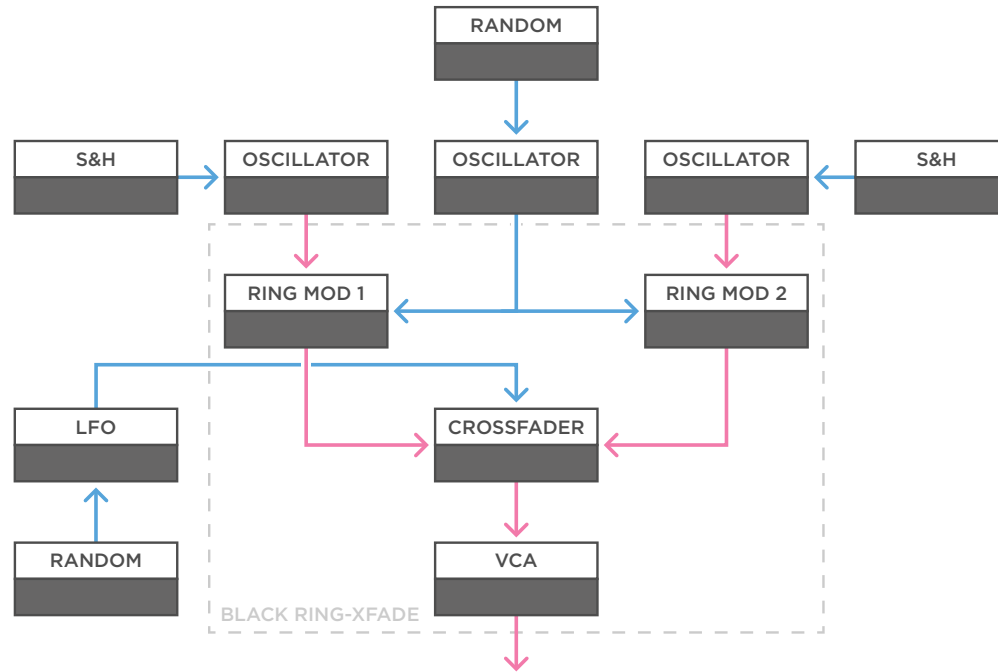
There's also a sequencer used to control the 1v/oct of the synth voice, as well as the modulation oscillator. And a sine wave LFO is used to modulate the cross-fader. Now, if you use the Ring CV control on the Black Ring-XFade to set the mod amount on the first ring mod to the max, and the second ring mod to none, the crossfader sweeps between a ring modulated melody, and a clean one.

Erica Synths Black Ring-XFade

Combined Patches

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - The Black Ring-XFade is also perfect for experimental soundscapes and drone patches. For example, you can use two oscillators on the inputs of ring mod one and two. And use another oscillator as the common modulation for both ring mods. Then add a slow LFO to modulate the crossfader.

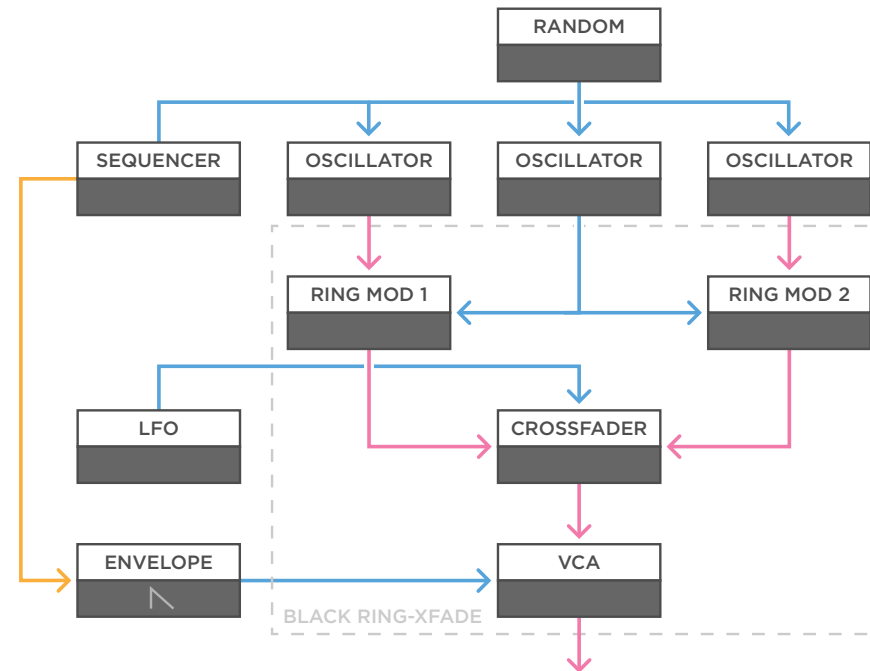
Add two tempo synced sample & hold circuits modulating the carrier oscillators, and a slow random voltage modulating the modulation oscillator. And finally, another slow random voltage modulating the LFO speed. You can get a wide range of haunting drone sounds and textures by experimenting with setups like this.

Erica Synths Black Ring-XFade

Combined Patches

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - In this patch there's a similar setup, with three oscillators, a random voltage to the modulator, and an LFO to the crossfader. But in this case the carrier inputs are modulated by a sequencer, to create a melody. The sequencer is also triggering a short percussive envelope, which in turn is used to control the internal VCA, for a more plucky melody.